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Dissemination of antibiotic resistance genes, mobile genetic elements, and efflux genes in anthropogenically impacted riverine environments

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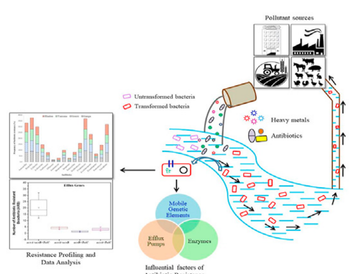
HIGHLIGHTS

- MGEs *intI1* and *intI2* were present in 89.25% and 1.07% in riverine isolates, respectively.
- Occurrence of ARGs and HMRGs were detected on MDR positive isolates in surface water.
- Efflux activity was confirmed in 96.26% isolates; with maximum presence of *acrA* 86.04%.
- Higher frequency of HMRGs (*zntB* 52.24%) was found in riverine isolates.

GRAPHICAL ABSTRACT



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ABSTRACT

Anthropogenically impacted surface waters are an important reservoir for multidrug-resistant bacteria and antibiotic-resistant genes. The present study aimed at MDR, ESBL, AmpC, efflux genes, and heavy metals resistance genes (HMRGs) in bacterial isolates from four Indian rivers belonging to different geo-climatic zones, by estimating the mode of resistance transmission exhibited by the resistant isolates. A total 71.27% isolates exhibited MDR trait, showing maximum resistance towards β-lactams (P = 66.49%; AMX = 59.04%), lincosamides (CD = 65.96%), glycopeptides (VAN = 25.19%; TEI = 56.91%), cephalosporins (CF = 53.72%; CXM = 30.32%) sulphonamide (COT = 43.62%; TRIM = 12.77%), followed by macrolide and tetracycline. The *dfrA1* and *dfrB* genes were detected in total 37.5% isolates whereas; *dfrA1* genes were detected in 33.34%. The *sul1* gene was detected in 9.76% and *sul2* gene was detected in 2.44% isolates. A total of 69.40% MDR integron positive isolates were detected with *intI1*and *intI2* detected at 89.25% and 1.07%, respectively; encoding class 1 and class 2 integron-integrase. ESBL production was confirmed in 73.13% isolates that harboured the genes *blaTEM* (96.84%), *blaSHV* (27.37%), *blaOXA* (13.68%)

Abbreviations: BOD, Biochemical Oxygen Demand; COD, Chemical Oxygen Demand; TDS, Total Dissolved Solid; TSS, Total Suspended Solid; DO, Dissolved oxygen; EC, Electrical Conductivity; PCR, Polymerase Chain Reaction; CFU, Colony-forming unit; MDR, Multidrug resistance; ARB, Antibiotic resistance bacteria; ARGs, Antibiotic resistance genes; ESBL, Extended-spectrum beta-lactamase; AST, Antibiotic sensitivity testing; AMR, Antimicrobial Resistance; HMRGs, Heavy metal resistance genes; MICs, Minimum inhibitory concentrations; AAS, Atomic Absorption Spectroscopy; MGEs, Mobile Genetic Elements BDL; Below detection limit RND, Resistance Nodulation Cell Division.

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Heavy metal resistance genes
Integrons

and *blaCTXM* (18.95%) while the frequency of HMRGs; 52.24% (*zntB*), 33.58% (*chrA*), and 6.72% (*cadD*). Efflux activity was confirmed in 96.26% isolates that harbored the genes *acrA* (93.02%), *tolC* (88.37%), and *acrB* (86.04%). AmpC (plasmid-mediated) was detected in 20.9% of the riverine isolates. Detection of such hidden molecular modes of antibiotic resistance in the rivers is alarming that requires urgent and stringent measures to control the resistance threats.

1. Introduction

Urbanization, industrialization, and human activity promote antibiotic-resistant bacteria into the environment, mainly in the aquatic system. Globally, antibiotic resistance is a major cause of healthcare-associated concerns, and in a wider population resistance has also emerged in infections. In the past few decades, the extensive application of anti-bacterial formulation in health care and veterinary sectors for treatment and farming have made antibiotic resistance is becoming an alarming threat (WHO, 2015; Robinson et al., 2016; Kumar and Pal., 2018; Hendriksen et al., 2019). Environment plays a role in the reservoir of ARGs, exposure to environmental antibiotic resistance rise due to the large population and disposal of wastes and wastewater in the riverine system (Sabri et al., 2018; Chaturvedi et al., 2020a & b). Antibiotic resistance has increased by extensive use and misuse of antibiotics and it has been globally accepted (Finley et al., 2013; Bell et al., 2014; Van et al., 2014). Due to the broad spectrum of antibiotics; MDR, ESBL, and AmpC producing bacterial isolates have increased day by day (Ye et al., 2017). Globally, 95% of the whole pool of drug distribution has been covered by an antibiotic like, beta-lactam and tetracycline (ECDC, 2015). Due to huge antibiotic contamination, environmental microbes like bacteria have evolved various mechanisms against antibiotics at a genetic and phenotypic level e.g. aggregation and biofilm formation (Brooks and Brooks, 2014; Corno et al., 2014). Bacteria being capable of fast mutation have developed a variety of mechanisms to resist heavy metals through natural selection of substrate contact with heavy metal ions (Poole, 2017). In the presence of heavy metals, the ARG selection system indirectly supports and maintains antibiotic resistance in the environment (Baker-Austin et al., 2006). Mobile genetic elements are frequently connected with ARGs, which makes them easily transferrable between bacterial species (Karkman et al., 2018; Gillings, 2014). Integrons are associated with gene cassettes that are responsible for the transmission of ARGs (Ravi et al., 2014). Sulphonamide resistance genes (*sul1*) and mobile genetic element (*int1*) acts as representative to monitor of ARGs (Narciso -da - Rocha et al., 2014; Chaturvedi et al., 2020a).

In horizontal gene transfer, various genetic elements participate in addition to transposons, conjugative plasmids, and integrons that help in retaining exogenously expressed genes (Uyaguari-Diaz et al., 2018). Previous studies have found ARGs with comparable diversities of *tet*(W), *bla*SRT-1, *erm*(B), *qnr*, *fts*H, and *mtr*AB in hospitals, and agriculture settings, respectively resistance to fluoroquinolones, tetracycline, beta-lactams, macrolides, aminoglycosides, and multidrug resistance (efflux pumps) (Hartmann et al., 2016; Fahimipour et al., 2018). Efflux pump helps to minimize the concentration of substrate within the cell and transfer into the intra to the extracellular environment. RND superfamily of the efflux pump is present in Gram-negative bacteria, mainly involved in the onset of the MDR, which spans the periplasm connecting the extracellular and intracellular membrane (Li et al., 2015; Yamaguchi et al., 2015). Molale and Bezuident (2016) performed a study to evaluate the presence of efflux genes (*mefA*, *tetK*, *tetL*, *msrC*) in *Enterococcus* spp. In which 17% presences of tetL and msrC have been identified. To date, various efflux pumps in pathogenic bacteria and *AcrAB-TolC* in *E. coli. MDR1*, *FLU1*, *CDR1*, and *CDR2* genes encoding efflux pump were found to influence by the presence of antibiotics in *Candida albicans* isolates in a water sample (Monapathi et al., 2017). Very few information is available concerning the occurrence or dissemination of ARGs in surface water (Chaturvedi et al., 2020b).

The novelty of this study is as follows: (i) environmental dissemination of emerging ARGs, efflux genes, and HMRGs (ii) MDR in anthropogenically impacted frontline Indian riverine system. Thus, this study was aimed to detect the dissemination of ARGs, heavy metal resistance genes (HMRGs), β-lactams, mobile genetic elements (*intI1*and *intI2*), and efflux genes in anthropogenically impacted surface waters of Indian rivers, viz. Ganga, Gomti, Yamuna, and Hindon belonging to different geo-climatic zones.

2. Material and methods

2.1. Sample collections

The surface water samples of river Ganga, Gomti, Yamuna, and Hindon were collected in 1 L sterilized glass bottles as per Bureau of Indian Standards (BIS, 2012) guidelines from different ecological conditions and geo-climatic zones of India. These sites are heavily loaded with several anthropogenic pollutions; samples were collected from 5 m away from the river bank at 12-inch depth from the water surface in triplicate (starting point, mid-point, and endpoint). Samples were collected at the end of December 2018. Collected samples were immediately kept in ice for transport to the laboratory for further analysis.

2.2. Surface water quality analysis

To test the quality of surface water physico-chemical parameters were analyzed as per the BIS (2012) and American Public Health Association methods (APHA, 2017). Different physico-chemical parameters like alkalinity, BOD, COD, TSS, TDS, sulfate, phosphate, and nitrate test were conducted as per BIS and APHA. Parameters like EC, temperature, and DO were analyzed on-site using multiparameter meter U-52 (Horiba, Japan). In surface water, heavy metals such as Ca, Mg, Ni, Mn, Cd, Cr, Zn, and Pb concentration was analyzed by using Atomic Absorption Spectroscopy (AAS) instrument (Pin AAcle 990 F, PerkinElmer, Waltham, Massachusetts, U.S.).

2.3. Isolation and screening of bacterial cultures

2.3.1. Bacterial isolation

The spread plate method was used for the isolation of culturable bacteria from the surface water samples (APHA, 2017). In sterile saline solution (0.9% NaCl) serial dilutions (10^{-1} to 10^{-6}) were prepared and inoculated into enriched Plate Count Agar Media (HiMedia, India) at 37 °C for 24–26 h for the quantitative analysis. The colonies were counted and isolates were purified based on morphological features by using Gram's reaction.

2.3.2. Screening of isolates

The bacterial isolates were tested for resistance and sensitivity towards different types of antibiotics which were assessed by the disc diffusion method by Kirby-Bauer 1961 (CLSI, 2017). Smyth et al. (2020) reported that isolates having resistance against three or more classes of antibiotics were considered as MDR bacterium. A total 10 different antimicrobial groups with 16 antibiotics were used; Cephalosporin- [Cefazoline (CZ; 30 μg), Cefuroxime (CXM; 30 μg)], Beta-lactam- [Penicillin (P; 30 μg), Amoxicillin (AMX; 30 μg)], Aminoglycosides- [Streptomycin (S; 10 μg), Gentamicin (GEN; 10 μg))], Fluoroquinolones- [Norfloxacin (NX; 10 μg), Ofloxacin (OF; 5 μg)] Carbapenems- [Imipenem (IPM; 750 μg)], Tetracyclin- [Tetracyclin (TET; 30 μg)], Lincosamides- [Clindamycin (CD; 2 μg)] Macrolides- [Erythromycin (E; 15 μg)] Sulphonamides- [Co-trimoxazole (COT; 25 μg), Trimethoprim (TRIM; 25 μg)], Glycopeptides- [Vancomycine (VAN; 30 μg) and Teicoplanin (TEI; 30 μg)].

As per the formulae given Antibiotic-resistant frequency was calculated by Su et al. (2012) as given below:

Antibiotic-resistant frequency (%) = (a/b) x 100

Where 'a' = no. of isolates resistant to antibiotics.

'b' = no. of isolates

2.3.3. Determination of minimum inhibitory concentration (MICs)

Heavy metals MICs were determined by the broth microdilution method (Wu et al., 2015). Isolates were tested for the heavy metal resistance in tryptic soy agar medium overnight at 37 °C. Four heavy metals were selected for MICs including cadmium (Cd), Zinc (Zn), Copper (Cu), and Chromium (Cr). The MICs concentration of the above four heavy metals are 30 mg/L −1600 mg/L. The range of concentration is used to analyze the MICs of heavy metal that completely inhibit the growth of isolates within 18−20 h.

2.4. Screening of β-lactamases (ESBLs and AmpC)

The Double-Disk Synergy Test (DDST) method was used for the bacterial isolates to test the enzymes e.g. ESBL and AmpC. ESBL production was tested by using Ceftazidime (CAZ 30 μg), Cefotaxime (CTX 30 μg), and Augmentin- AMC (20 μg Amoxicillin and 10 μg Clavulanic acid) disk used for the confirmatory test (CLSI, 2013). A positive result was confirmed based on zone expansion on the augmentin disk. The presence of AmpC was observed by using Cefoxitin (CX 30 μg) and Cefoxitin-Cloxacillin (CXX 200 μg); the zone increased with a diameter of ≥ 4 mm in the CXX zone than CX alone indicates a positive result (Bajaj et al., 2015).

2.5. Molecular detection of ARGs and HMRGs in riverine isolates

The alkaline lysis method was used for the genomic DNA extraction of MDR positive bacterial isolates. The genomic DNA purity and concentration were analyzed using a spectrophotometer (ND-1000, Thermo Scientific, US) and agarose gel electrophoresis (Takara mupid 12269, Japan). PCR (Eppendorf AG, Germany) was used to observe the existence of sulfonamide (co-trimoxazole and trimethoprim) determinants (*sul1, sul2,* and *dfrA1, dfrB*) (Maynard et al., 2003; Mohamed et al., 2017) (Supplementary Information: S1). The detection of HMRGs e.g. *cadD, zntB, pcoA,* and *chrA* were analyzed by PCR using a set of primers (Yang et al., 2020; Argudin et al., 2016) (Supplementary Information: S1).

2.6. Detection of integrons

In MDR positive isolates, PCR was used to check the presence of integron-integrase genes *intI1* and *intI2* (Mazel et al., 2000) (Supplementary Information: S1). During the horizontal gel electrophoresis, a 100-bp DNA standard (New England Biolabs, USA) was used as a marker. The amplicons were visualized under UA transilluminator (G: BOX Syngene, USA).

2.7. Detection of beta-lactamase and efflux genes

To detect the *blaCTXM (CTXM1, CTXM2,* and *CTXM9)* and *blaTEM, blaSHV, blaOXA*, and genes, multiplex PCR was used as per the conditions described by Dallenne et al. (2010) (Supplementary Information: S1). Further, it was also used for the detection of *blaCIT, blaDHA, blaACC, blaEBC, blaMOX,* and *blaFOX*. As per the method described by Pérez and Hanson (2002), the detection of AmpC genes were observed using multiplex PCR. 100-bp and 1-kb DNA marker (New England Biolabs, USA) as standard, and products were visualized under UA transilluminator (G: BOX Syngene, USA). The presence of efflux genes (*acrA, acrB*, and *tolC*) and tetracycline genes (*tetA*) were analyzed by PCR (Kafilzadeh and Farsimadan, 2016; Maynard et al., 2003) (Supplementary Information: S1).

2.8. Statistical analysis

Physico-chemical parameters, ARGs, and efflux genes were analyzed for Pearson's correlation coefficient. Besides, the association between antibiotic resistance, antibiotic resistance genes as well as efflux pump were also analyzed for Pearson's correlation. A *t*-test was performed for the relationship of Antibiotic resistance and its associated genes.

3. Results and discussions

3.1. Water quality of the river system

Table 1 shows the results of the river water samples values (mean ± SD). The water quality of rivers (R1, R2, R3, and R4) was affected considerably. The COD and TSS of Ganga were found higher 108 mg/L and 527 mg/L, respectively in comparison to other rivers. Similarly, Nwankwo and Ovunda (2019) reported that physico-chemical values range like, pH (7.02−8.50), chloride (1008−1991 mg/L), and TDS (3312−7566 mg/L) in a river water sample. Chatanga et al. (2019) have also reported the physico-chemical values of river water significantly changes beyond the expected limits, except pH. The finding indicates that river water is severely affected by agriculture, industries, and various other anthropogenic activities. Loucif et al. (2020) carried out a study on the surface water of a lake, which showed an average pH of 7.2, DO 0.70 mg/L, EC 1747.3 μS/cm, NO_3^- 17.5 mg/L, and BOD_5 3.8 mg/L. Aniyikaiye et al. (2019) reported in a paint industry effluents the values in the range of pH (4−12.2), EC (149.1−881.3 μS/m), and TDS (1100−6510 mg/L) respectively. The range of other parameters include TSS (0−2470 mg/L), Chloride (63.8−733.8 mg/L), DO (0−6.7 mg/L), BOD (162.8−974.7 mg/L), COD (543−1231 mg/L), phosphate (BDL-0.02 mg/L), Nitrates (12.89−211.2 mg/L), Nickel (BDL-1.9 mg/L), Sulfate (195−1434 mg/L) while copper, lead and chromium were below detection limits. Friedl et al. (2004) and UNEP GEMS/Water Programme (2006) reported that the pH values between 6.5 and 8.5 indicated good water quality. Chatanga et al. (2019) reported that the acid-neutralizing capacity of the water was reduced due to the high pH of the water that fluctuated along its course with the presence of acid-forming PO_4^{3-} and NO_3^-.

Table 1
Water quality analysis (Physico-chemical and metals) of the studied rivers.

Physico-chemical Parameters Parameters	Ganga (R1)	Gomti (R2)	Yamuna (R3)	Hindon (R4)
pH	7.52 ± 0.20	7.67 ± 0.15	7.68 ± 0.09	7.58 ± 0.17
DO (mg/L)	5.20 ± 0.06	5.03 ± 0.15	5.80 ± 0.20	5.10 ± 0.20
Temp (∘C)	25 ± 0.70	29 ± 0.35	25 ± 1.00	25 ± 0.57
EC (μS/cm)	284 ± 4.50	652 ± 2.00	347 ± 2.08	240 ± 1.73
BOD (mg/L)	30 ± 2.83	19 ± 0.58	14 ± 1.41	15 ± 0.70
COD (mg/L)	108 ± 2.00	72 ± 2.00	43 ± 0.02	39 ± 2.51
TSS (mg/L)	126 ± 1.53	527 ± 6.00	81 ± 1.00	85 ± 1.00
TDS (mg/L)	192 ± 0.58	381 ± 1.00	239 ± 1.00	146 ± 1.52
Alkalinity (mg/L)	151 ± 1.15	269 ± 1.00	161 ± 1.00	130 ± 0.00
SO_4^{2-} (mg/L)	10.23 ± 0.59	10.01 ± 0.52	5.33 ± 0.01	6.90 ± 0.36
PO^{4-}(mg/L)	0.43 ± 0.01	0.30 ± 0.00	0.75 ± 0.01	0.26 ± 0.21
NO^{3-}(mg/L)	0.72 ± 0.03	0.14 ± 0.01	0.02 ± 0.01	0.16 ± 0.01
Metal Analysis Parameters	**Ganga (R1)**	**Gomti (R2)**	**Yamuna (R3)**	**Hindon (R4)**
Cd (mg/L)	0.002 ± 0.01	0.003 ± 0.00	0.002 ± 0.00	0.001 ± 0.00
Cr (mg/L)	BDL	0.335 ± 0.00	BDL	BDL
Pb (mg/L)	BDL	0.001 ± 0.00	0.005 ± 0.00	0.002 ± 0.00
Zn (mg/L)	0.0033 ± 0.00	0.046 ± 0.00	BDL	0.503 ± 0.00
Ni (mg/L)	BDL	0.002 ± 0.00	BDL	0.001 ± 0.00
Ca (mg/L)	34.700 ± 0.15	48.400 ± 0.20	40.8 ± 0.05	40.300 ± 0.58
Mg (mg/L)	3.285 ± 0.06	11.537 ± 0.01	3.320 ± 0.07	11.423 ± 0.23
Mn (mg/L)	0.0108 ± 0.00	0.047 ± 0.01	0.041 ± 0.00	0.013 ± 0.00

R1: Ganga river; R2: Gomti river; R3: Yamuna river; R4: Hindon river; BDL: below detection limit; EC: Electric Conductivity.

3.2. Enumeration of bacterial isolates

The heterotrophic isolates ranged from 4.7×10^4 to 6.1×10^6 CFU/mL in Ganga, 5.1×10^5 to 4.7×10^6 CFU/mL in Gomti, 5.5×10^4 to 4.0×10^6 CFU/mL in the Yamuna, and 4.2×10^4 to 3.9×10^6 CFU/mL in Hindon. There has been variation in the numbers of bacterial species and counts in different rivers and other sources of water such as lakes as reported by various researchers, which has been primarily linked with anthropogenic and related activities. Nwankwo and Ovunda (2019) reported

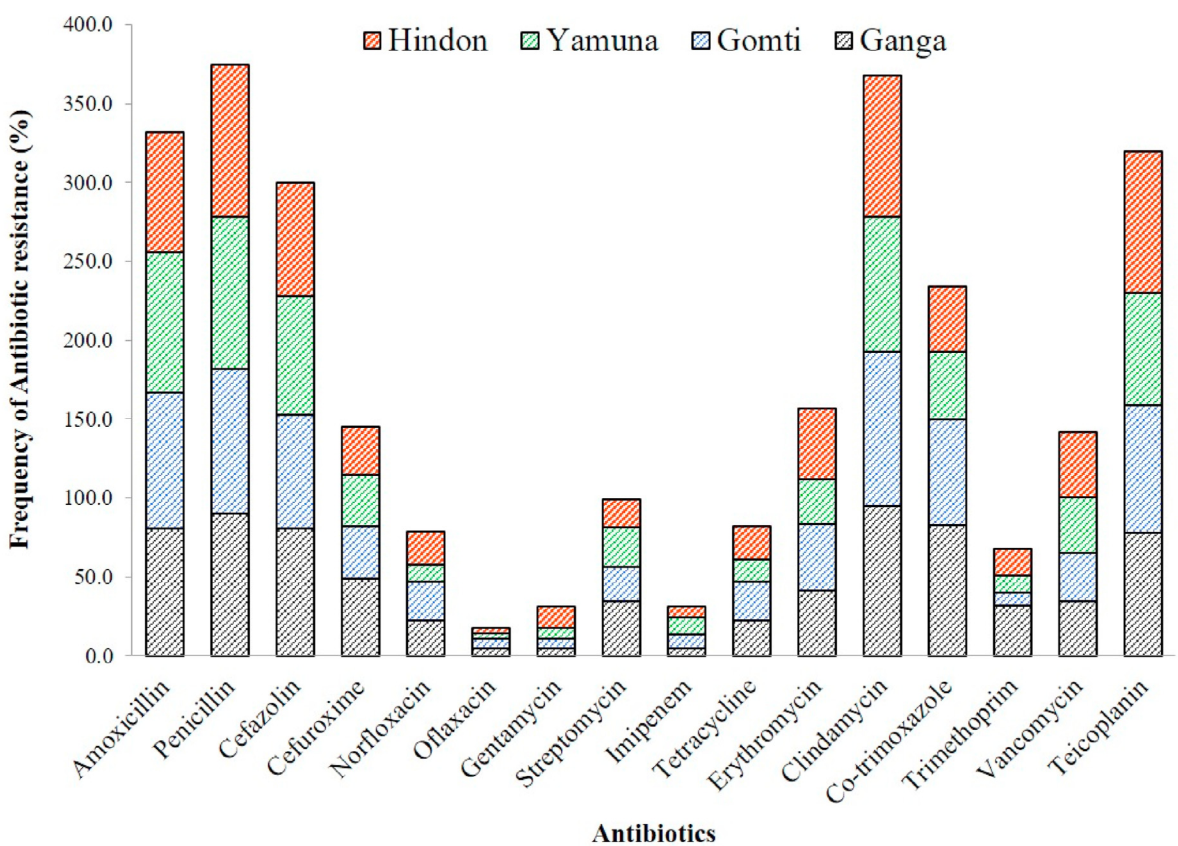

Fig. 1. Distribution of antibiotic resistant in (R1-Ganga, R2-Gomti, R3-Yamuna, and R4-Hindon) riverine systems.

Table 2
Antibiotic resistance profiling and distribution of integrase genes in MDR positive isolates.

Isolates	Antibiotic resistance class	Efflux pump				Antibiotic genes detected	Heavy Metal resistance genes				Beta lactam genes	integrons
		RND			Tetracycline							
		acrA	acrB	tolC	tetA		cadD	zntB	pcoA	chrA		
R1-AT-1	β-lactam, Cephalosporin Lincosamide, Sulfonamide, Glycopeptide	+	+	+		*dfrA1,dfrB, sul1, sul2*	−	+	-	+	*blaTEM, blaCTXM1, blaCTXM9*	*intI1, intI2*
R1-AT-5	β-lactam, Cephalosporin, Floroquinolones, Aminoglycoside, Lincosamide, Sulfonamide, Glycopeptide	+	+	+		*sul1*	+	+	+	-	*blaTEM, blaMOX, blaDHA*	*intI1*
R1-AT-8	β-lactam, Cephalosporin, Tetracycline, Macrolide, Sulfonamide	+	+	+	+	*dfrA1, dfrB,*	−	+	+	+	*blaTEM*	*intI1*
R1-AT-11	β-lactam, Cephalosporin, Macrolide, Lincosamide, Sulfonamide, Glycopeptide	+	+	+		*sul1, sul2*	+	+	-	+	*blaTEM, blaSHV, blaOXA*	*intI1, intI2*
R1-AT-15	β-lactam, Cephalosporin, Macrolide, Lincosamide, Sulfonamide, Glycopeptide	+	+	−		*sul1, sul2*	−	+	+	+	*blaTEM, blaSHV, blaOXA*	*intI1*
R1-AT-26	β-lactam, Cephalosporin, Aminoglycoside, Lincosamide, Sulfonamide, Glycopeptide	+	+	+		*dfrA1,*	−	+	+	+	*blaTEM*	*intI1*
R1-AT-27	β-lactam, Cephalosporin, Floroquinolones, Lincosamide, Sulfonamide, Glycopeptide	+	+	+		*dfrA1,dfrB, sul1,*	−	+	-	+	*blaTEM, blaCTXM1*	*intI1*
R1-AT-31	β-lactam, Cephalosporin, Floroquinolones, Tetracycline, Lincosamide, Sulfonamide, Glycopeptide	+	+	+	+	*dfrA1,dfrB, sul1, sul2*	−	+	-	+	*blaTEM, blaSHV*	*intI2*
R1-AT-34	β-lactam, Cephalosporin, Tetracycline, Lincosamide, Sulfonamide	+	+	+	−	*sul1, sul2*	+	-	-	-	*blaTEM, blaSHV, blaOXA*	*intI1, intI2*
R1-AT-37	β-lactam, Floroquinolones, Aminoglycoside, Lincosamide, Sulfonamide, Glycopeptide	+	+	+		*sul1, sul2*	−	+	-	-	*blaTEM, blaCTXM1, blaCTXM9*	*intI1*
R1-AT-38	β-lactam, Cephalosporin, Lincosamide, Sulfonamide, Glycopeptide	+	+	+		*dfrA1, sul1, sul2*	-	+	+	+	*blaTEM*	*intI1*
R2-AT-2	β-lactam, Cephalosporin, Floroquinolones, Carbapenem, Macrolide, Lincosamide, Sulfonamide, Glycopeptide	+	+	+		*dfrA1, sul1, sul2*	−	+	+	-	*blaCTXM9*	*intI1*
R2-AT-8	β-lactam, Cephalosporin, Lincosamide	+	+	+			-	+	+	-	*blaTEM*	*intI1*
R2-AT-11	β-lactam, Floroquinolones, Lincosamide, Sulfonamide, Glycopeptide	+	+	+		*sul1, sul2*	+	+	-	+	*blaTEM, blaCTXM1*	*intI1*
R2-AT-14	β-lactam, Cephalosporin Tetracycline, Lincosamide, Sulfonamide, Glycopeptide	+	+	+	+	*sul1, sul2*	-	+	+	-	*blaTEM*	*intI1*
R2-AT-20	β-lactam, Macrolide, Lincosamide, Sulfonamide, Glycopeptide	+	−	−		*sul1,*	−	+	-	+	*blaTEM, blaSHV, blaCTXM2*	*intI1*
R2-AT-22	β-lactam, Lincosamide, Sulfonamide, Glycopeptide	+	+	+		*sul1, sul2*	-	-	+	-	*blaTEM*	*intI1*
R2-AT-25	β-lactam, Cephalosporin, Aminoglycoside, Lincosamide, Sulfonamide	+	+	+		*sul1, sul2*	-	+	-	-	*blaTEM, blaACC*	*intI1*
R2-AT-29	β-lactam, Cephalosporin, Tetracycline, Macrolide, Lincosamide, Glycopeptide	−	−	−	+		-	+	-	-	*blaTEM, blaEBC, blaACC*	*intI1*
R2-AT-31	β-lactam, Cephalosporin, Macrolide, Lincosamide, Sulfonamide, Glycopeptide	+	+	−		*sul1, sul2*	+	+	-	+	*blaTEM*	*intI1*
R3-AT-11	β-lactam, Lincosamide, Sulfonamide, Glycopeptide	+	+	+			+	+	-	-	*blaTEM, blaSHV*	*intI1*
R3-AT-18	β-lactam, Cephalosporin, Lincosamide, Sulfonamide, Glycopeptide	−	+	−		*sul1, sul2*	+	+	-	+	*blaTEM*	*intI1, intI2*
R3-AT-22	β-lactam, Cephalosporin, Aminoglycoside, Carbapenem, Sulfonamide, Glycopeptide	+	+	+		*sul1, sul2*	+	-	+	+	*blaTEM, blaSHV*	*intI1*
R4-AT-1	β-lactam, Lincosamide, Sulfonamide, Glycopeptide,	+	+	+		*sul1, sul2*	-	+	-	+	*blaTEM, blaSHV, laOXA*	*intI1*
R4-AT-2	β-lactam, Cephalosporin, Floroquinolones, Lincosamide, Glycopeptide,	+	+	+		*sul1, sul2*	-	+	+	-	*blaTEM, blaSHV, blaOXA*	*intI1, intI2*
R4-AT-9	β-lactam, Cephalosporin, Macrolide, Lincosamide, Sulfonamide, Glycopeptide	+	+	−		*sul1, sul2*	-	-	-	+	*blaTEM*	*intI1*
R4-AT-10	β-lactam, Macrolide, Lincosamide, Sulfonamide, Glycopeptide	+	+	+		*sul1, sul2*	-	+	-	+	*blaTEM*	*intI1*
R4-AT-24	β-lactam, Cephalosporin, Tetracycline, Lincosamide, Sulfonamide, Glycopeptide	+	+	+	−	*dfrA1, dfrB,*	-	+	+	-	*blaTEM, blaSHV, blaOXA, blaCIT*	*intI1*
R4-AT-28	β-lactam, Aminoglycoside, Lincosamide,	+	+	+			-	+	-	+	*blaTEM, blaOXA*	*intI1*

+: Positive; -: Negative; Sulphonamide genes: (*sul1, sul2*); Trimethroprim genes (*dfrA1, dfrB*); tetracycline genes (*tetA*); RND (Resistance Nodulation Cell Division) efflux genes (*acrA, acrB*, and *tolC*).

heterotrophic bacterial count in the river from 14.0×10^4 CFU/mL to 100.0×10^4 CFU/mL. Chaturvedi et al. (2020) reported the CFU/mL count of bacteria in four rivers in northern and southern India as 2.3×10^5 to 5.7×10^6 CFU/mL in Ganga, 1.9×10^5 to 4.2×10^6 CFU/mL in Gomti, 1.2×10^5 to 3.9×10^6 CFU/mL in Narmada, and 2.1×10^5 to 3.5×10^6 CFU/mL in Kaveri.

3.3. Distribution pattern of the riverine system

A total of 188 bacteria were isolated from four riverine systems. During MDR screening a total of 16 antibiotics (10 different groups) were used. A total of 134 (71.27%) isolates showed the MDR trait i.e., resistant to three or more than three classes of antibiotics

(Supplementary Information: S2). The beta-lactams showed maximum resistance (P = 66.49%; AMX = 59.04%), lincosamide (CD = 65.96%), glycopeptide (VAN = 25.19%; TEI = 56.91%), cephalosporin (CF = 53.72%; CXM = 30.32%) followed by sulfonamide (COT = 43.62%; TRIM = 12.77%), macrolide (ER = 28.19%) and tetracycline (TET = 14.89%), fluoroquinolones (NX = 14.36%; OF = 3.19%), aminoglycosides (GEN = 5.32%; S = 18.09%) and carbapenem (IPM = 5.32%) showed least resistance (Fig. 1). Koczura et al. (2012) studied the antibiotic resistance in river (upstream and downstream) water, which was as follows: streptomycin (100% and 97.8%), tetracycline (61.1% and 82.2%), norfloxacin (5.6% and 55.6%) and co-trimoxazole (55.6% and 88.9%). Su et al. (2012) reported the antibiotic resistance profile of *E. coli* strains in Dongjiang River, South China. In this study 12 antibiotics are tested in which tetracycline found in highest resistance frequency (TET = 60.1%) followed by streptomycin (S = 56.3%), ampicillin (AMP = 43.3%), trimethoprim (TRIM = 41.6%), piperacillin (PIP = 35.0%), chloramphenicol (CHL = 27.5%), ciprofloxacin (CIP = 18.9%), gentamicin (GEN = 16.7%) and levofloxacin (LEV = 14.9%), caphazolin (CZ = 10.2%) and ceftazidime (CAZ = 2.7%).

3.4. Prevalence of MICs in riverine isolates

MICs of heavy metals for surface water isolates range from 30 mg/L to 1500 mg/L. The MIC values for heavy metals were Cr (350–800 mg/L), Zn (400–1450 mg/L), Cu (750–1500 mg/L), and Cd (30–450 mg/L). Yang et al. (2020) reported that MICs of metals were as follows: ≤12.5–25 mg/L (Hg), 400–800 mg/L (Cr), 400–1600 mg/L (Zn), 100–200 mg/L (Co), 800–1600 mg/L (Cu), 800–1600 mg/L (Mn), and 50–800 mg/L (Cd). Rahman and Singh (2018) reported that MICs of Heavy metals used in different concentrations included 50–1600 mg/L (Cr(VI)) and (Cd), 50–3200 mg/L (Cu), 50–1200 mg/L (As), 50–4500 mg/L (Pb) and 50–100 mg/L (Hg). A total of 90 *V. parahaemolyticus* isolates were analyzed for their susceptibility to heavy metals. As compared to the *E. coli* K-12 C600 quality control strain, the maximum MIC values were observed of 3200 μg/mL for Cu^{2+}, Cd^{2+}, Ni^{2+}, and Zn^{2+} as well as 1600 μg/mL for Cr^{3+} and Co^{2+} in a study (Jiang et al., 2020).

3.5. Determination of ARGs and HMRGs

Out of 134 MDR positive isolates, 17.91% of isolates show trimethoprim resistance. The *dfrA1* genes were detected in eight (33.34%) and *dfrA1-dfrB* genes were detected in nine (37.5%). 61.19% isolates show Co-trimoxazole resistance. the *sul1* gene was detected in six (9.76%) isolates and *sul2* gene was detected in two (2.44%). Both *sul1* and *sul2* genes were detected in 52 (63.41%) (Table 2) (Supplementary Information: S2). In a research study, *sul1* and *sul2* were detected i.e., 21.6% and 27.8% in a drinking water sample (Adesoji et al., 2017). In another study *sul1* and *sul2* reported from surface water were 2.6% and 1.6% (Makowska et al., 2016). Four HMRGs were detected among the riverine isolates including *cadD, zntB, pcoA, chrA*. The frequency of HMRGs as follows *zntB* (52.24%), *chrA* (33.58%), *pcoA* (21.64%) and *cadD* (6.72%). Yang et al. (2020) reported 81.1% *zntB* and 77.0% *zntA* in *Salmonella* and *E.coli* isolates from chicken farms and in Salmonella species from retail meat *zntB* frequency was high (92.0%) as compare to *zntA* (0.6%).

3.6. Detection of intI1 and intI2 by PCR

A total of 93 (69.40%) MDR positive isolates were screened for *intI1* and *intI2* encoding class 1 and class 2 integron-integrase genes. *IntI1* gene was detected in 83 (89.25%) isolates, whereas *intI2* was in one (1.07%) isolate. A total of nine (9.68%) isolates carried both integron-integrase genes (*intI1* and *intI2*) (Supplementary Information: S2). Adesoji et al. (2017) reported these values as 21.42% and 3.6% in class 1 and class 2, respectively in the drinking water. Chaturvedi et al. (2020b) recently report mobile genetic elements in the riverine system i.e., 60.3% of isolates show *IntI1* gene, and *intI2* was observed in 9.09% of isolates. 6.61% isolates were also carried both *intI1* and *intI2* genes in this study. Among the 120 *E. coli* strains, 65.0% of isolates had the integrons genes. The *intI1* (61.7%) and *intI2* (6.7%) genes were detected in strains respectively, however, the *intI3* gene was not detected in any of the isolates. 3.3% isolates show both *intI1* and *intI2* genes (Alkhudhairy et al., 2019).

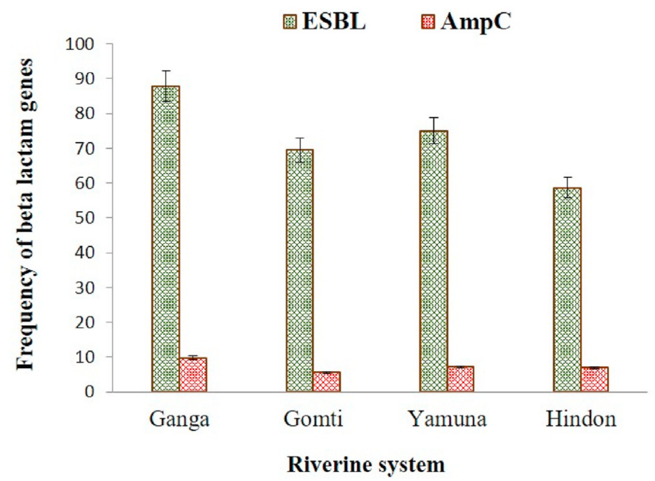

Fig. 2. Prevalence of ESBL and AmpC in (R1-Ganga, R2-Gomti, R3-Yamuna, R4-Hindon) riverine systems.

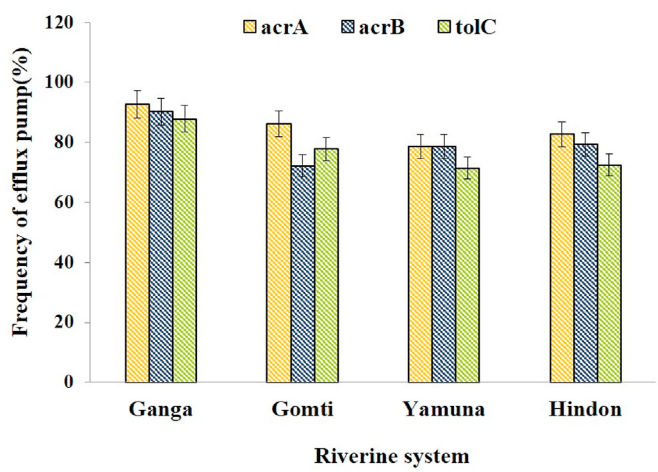

Fig. 3. Profiling of Efflux Pump in environment isolates from Ganga (R1), Gomti (R2), Yamuna (R3) and Hindon (R4).

3.7. Characterization of beta-lactamase production

From 134 MDR positive isolates, 98 (73.13%) were confirmed as ESBL producers. *blaTEM* (96.84%) was the most prevalent ESBL gene that was detected most of the isolates (ESBL positive) followed by *blaSHV* (27.37%), *blaOXA* (13.68%) and *blaCTXM* (18.95%) (Supplementary Information: S2). *blaTEM* was majorly (52.63%) reported ESBL gene. Some isolates harbored different combinations of ESBL genes (*blaTEM* + *blaSHV* = 13.68%;

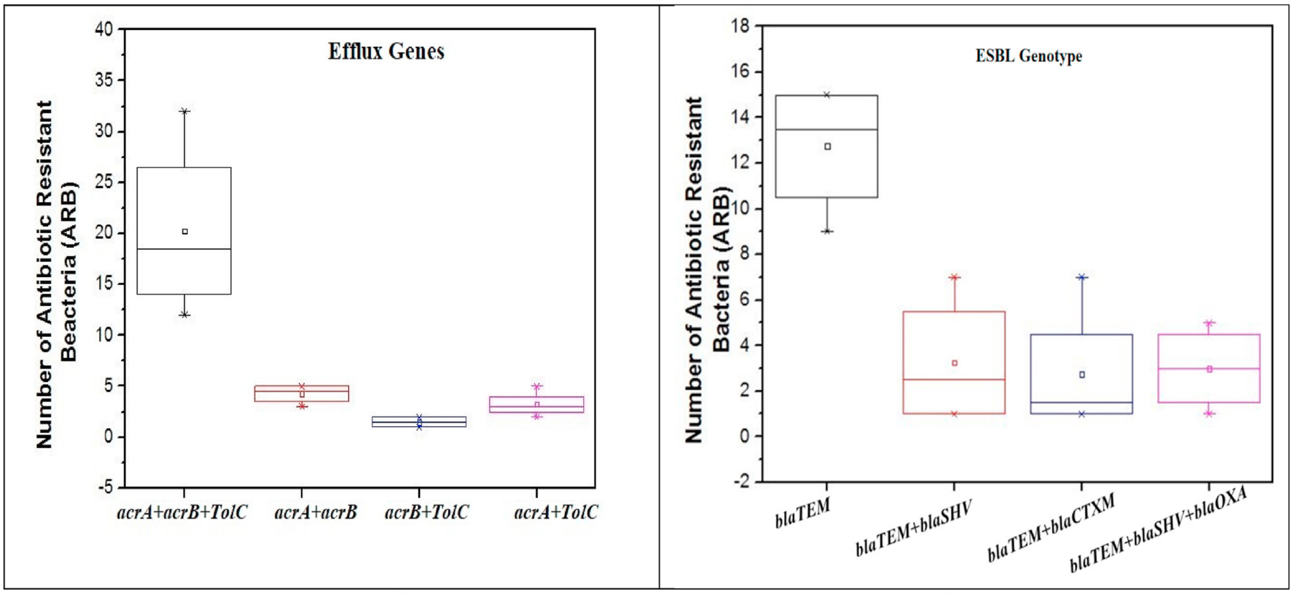

Fig. 4. Box plot showing presence of A: antimicrobial resistance (ARB) versus efflux genes and B: antimicrobial resistance (ARB) versus different ESBL genotypes in isolates. Solid horizontal line represents the median, the stems show the minimum and maximum values, and the cross indicates an outlier.

Table 3
Correlation coefficient of physico-chemical parameters, ARGs and MDR efflux Activity.

Correlation coefficient	pH	DO	Tem	EC	BOD	COD	TDS	*sul1*	*sul2*	*intI1*	*intI2*	*acrA*	*acrB*	*tolC*
pH	1.00													
DO	0.42	1.00												
Tem	0.50	−0.48	1.00											
EC	0.63	−0.27	0.97	1.00										
BOD	−0.73	−0.36	−0.05	−0.08	1.00									
COD	−0.58	−0.35	0.14	0.12	0.98	1.00								
TDS	0.67	−0.14	0.93	0.99	−0.07	0.14	1.00							
sul1	−0.49	−0.47	0.32	0.28	0.93	0.98	0.29	1.00						
sul2	−0.51	−0.46	0.29	0.26	0.94	0.99	0.26	1.00	1.00					
intI1	−0.52	−0.74	0.43	0.33	0.84	0.88	0.29	0.94	0.93	1.00				
intI2	−0.81	−0.16	−0.33	−0.35	0.96	0.89	−0.31	0.79	0.81	0.65	1.00			
acrA	−0.66	−0.49	0.14	0.08	0.98	0.99	0.08	0.98	0.98	0.93	0.88	1.00		
acrB	−0.76	−0.33	−0.10	−0.13	1.00	0.97	−0.11	0.91	0.92	0.82	0.97	0.97	1.00	
tolC	−0.66	−0.42	0.09	0.05	0.99	0.99	0.06	0.97	0.98	0.90	0.91	1.00	0.98	1.00

blaTEM + *blaSHV* + *blaOXA* = 12.63% *blaTEM* + *blaCTXM* = 12.63%). Chen et al. (2010) reported 96.9% *bla* genes (*blaTEM*, *blaCTXM*, *blaSHV* and *blaOXA*) and 30.4% *blaTEM* + *blaCTXM* present in Yangtze river china, they also consider *blaSHV* with *blaCTXM* (9.1%) as well as *blaTEM* with *blaSHV* (6.0%).

In surface water, some isolates were found positive for the plasmid-mediated beta-lactamase enzyme AmpC. A total of 20.90% out of 134 isolates were positive for AmpC production. While the frequency of AmpC positive isolates were reduced (7.46%) during the genotypic analysis. The AmpC genes that were identified in 10 isolates included *blaACC* (60.00%), *blaDHA* (30.00%), *blaCIT* (30%), *blaFOX* (20%), *blaEBC* (20.00%) and *blaMOX* (10.00%) (Fig. 4). Fig. 2 shows the distribution of ESBL and AmpC. Amador et al. (2015) reported ESBL (51.9%) and AmpC (44.4%) in Enterobacteriaceae in the wastewater sample, which included AmpC producer's *blaEBC* (38.9%), *blaFOX* (1.9%) and *blaCIT* (1.9%) reported.

3.8. Detection of efflux genes by PCR

The efflux genes (*acrA, acrB,* and *tolC)* and tetracycline efflux genes *(tetA)* were detected in MDR positive isolates using PCR Fig. 3. Efflux pump (*acrA, acrB,* and *tolC)* was detected in 129 (96.26%) isolates, whereas *acrA and acrB* was found in 15 (11.62%) isolates. In the present study, 93.02%, 88.37% and 86.04% of isolates had genes *acrA, tolC*, and *acrB* respectively (Supplementary Information: S2). Kafilzadeh and Farsimadan (2016) reported 51.1% *acrA,* 75.0% *acrB* and 69.4% *tolC* genes in *E.coli* strains from the patients in Iran. The frequency of the *acrA, acrB*, and *acrA* + *acrB* genes were 82.9%, 82.9%, and 95.9%, respectively reported in uropathogenic *E.coli* isolates (Maleki et al., 2016). Tetracycline resistance was detected in 29 (21.64%) isolates. Tetracycline efflux genes were detected in 20 (68.97%) isolates.

3.9. Data analysis

Pearson's Correlation coefficient among physico-chemical parameters and ARGs and efflux genes were shows in riverine system. The relationship between physico-chemical parameters and antibiotic resistance associated genes were significant positive and negative correlations. Positive correlation including to positive pairs such as BOD-*sul1* (R = 0.93), BOD-*sul2* (R = 0.94), BOD-*acrB* (R = 1.00), BOD-*tolC* (R = 0.99), BOD-*int1* (R = 0.84), BOD-*int2*

(0.96), COD-*sul1* (R = 0.93), COD-*sul2* (R = 0.99), COD-*acrA* (R = 0.99), COD-*acrB* (R = 0.97), COD-*tolC* (R = 0.99), COD-*int1* (R = 0.88) COD-*int2* (R = 0.89), respectively (Table 3). In contrast the negative correlation affected such as pH-*sul1* (−0.49), pH-*sul2* (−0.51), pH = *int1* (−0.52), pH = *int2* (−0.81), pH = *acrA* (−0.66), pH = *acrB* (−0.76), pH = *tolC* (−0.66) and DO is also negatively correlated with the antibiotic resistance associated genes. Pearson's correlation was also performed for the association of antibiotic resistance and its associated genes, which revealed a strong correlation between them e.g. AR-*sul1* (R = 0.99, P = 0.003), AR-*sul2* (R = 0.99, P = 0.002) and AR-*int1* (R = 0.95, P = 0.050). Positive correlations between the physicochemical parameters of Tonga Lake water included the following pairs like pH-NO_3^- (P = 0.015), pH-Cl^- (P = 0.024), EC-dry residues (P = 0.026), EC-Cl^- (P = 0.019), EC-NO_3^- (P = 0.001), NO_3^--Mg^{2+} (P = 0.043), NO_3^--Cl^- (P = 0.002), PO_4^{3-} -K^+(P = 0.000), Mg^{2+} −SO_2^{2-} (P = 0.009) and the negative correlations shows between Ca^{2+}-BOD_5 (P = 0.033), Mg^{2+}-BOD_5 (P = 0.010) and BOD_5-hardness (P = 0.019)(Loucif et al., 2020).

4. Conclusions

The involvement of different modes of transport of resistance determinants in the environment is critical and generates an urgent need for immediate measures to minimize the spread of this global menace. This study explains the dissemination of MDR, ESBL, AmpC β-lactamase, HMRGs, and efflux genes producing isolates in the Indian riverine system. The detection of isolates with resistant emergent genes generates an alarming situation for the public governing bodies to set up immediate measures for improving the microbiological quality of the river water. Furthermore, polluted water is the potent reservoir and a hot spot for the dissemination of many pathogens; the effluents thus released from tertiary health care, industries, and domestic sewage to be monitored routinely; before its release into the open environment for reuse. The resistant strains can easily get biomagnified and introduced into the food chain thereby causing infections and affecting the socio-economic status and has long-term negative impacts on the environment.

Credit authors statement

Preet Chaturvedi (PC) Methodology, experimentation work, Writing, Resources Conceptualization, Investigation, Writing − original draft preparation, Project administration, Funding acquisition. Pankaj Chowdhary (PC), Deepshi Chaurasia (DC) and Anuradha Singh (AS): Sampling, experimentation work (AMR screening), Methodology, Writing − original draft preparation. Ram Chandra, Ashok Pandey, and Pratima Gupta (PG): Supervision Formal analysis, Writing - Reviewing and Editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.chemosphere.2021.129693.

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